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Abstract

Sammendrag

Contents

Abstract	1
Sammendrag	2
Abbreviations	5
1 Introduction	6
1.1 Motivation	6
1.2 Limitations	6
1.3 Contributions	6
1.4 Thesis Structure	6
2 Literature Study	8
2.1 Introduction to Object Detection with Deep Learning Models	8
2.2 Introduction to Deep Learning Model Optimization	8
2.3 Yolo8 Model Architecture	8
2.4 Performance Challenges in Machine Learning Deployment	9
2.5 DNNs Compiler Workflow	12
2.6 Performance Bottlenecks in ML Inference	13
2.7 Hailo Compiler	14
2.8 Methods for Pruning Deep Neural Network	14
2.8.1 Magnitude based pruning	16
3 Theory	21
4 Methodology	22
5 Proposed Solution / Design	23
5.1 Roofline model	23
5.2 GPU structure	28
5.3 Apache TVM	29
5.4 Dataset and Annotation	29
6 Testing and Results	30
6.1 Experimental Setup	30

6.1.1	Performance overview	30
6.1.2	Prediction Examples	30
6.1.3	Hailo impact	30
6.2	FP32 Runtime Comparison on Jetson Orin	30
6.3	Quantization Influence on YOLOv8n	31
6.3.1	TensorRT FP32 vs INT8 Benchmarking on Jetson Orin	31
6.4	YOLO pruning	33
7	Discussion	34
7.1	Discussion Points	34
7.1.1	Quantization and accelerator deployment	34
7.1.2	Unexplored hardware configurations and deployment possibilities	34
8	Conclusion	35
9	Future Work	36
10	Description of Use of Artificial Intelligence	37
A	Demonstration (Appendix A)	41
B	Demonstration (Appendix B)	45

Abbreviations

1 Introduction

1.1 Motivation

1.2 Limitations

1.3 Contributions

Within these limitations, the main contributions of this thesis are:

- :
- :
- :
- :
- :

1.4 Thesis Structure

The remainder of this report is organised as follows:

- Chapter 2
- Chapter 3
- Chapter 4
- Chapter 5
- Chapter 6
- Chapter 7
- Chapter 8
- Chapter 9

- **Chapter 10**
- **Appendix A**
- **Appendix B**

2 Literature Study

2.1 Introduction to Object Detection with Deep Learning Models

Artificial Intelligence (AI) is a broad term covering a set of algorithms and operations on matrices and vectors that enable computers to perform a big variety of tasks in language understanding and translation, object recognition, robotic performance etc. We will consider only models performing object detection tasks, mostly having CNN [1] and Transformer [2] architectures.

2.2 Introduction to Deep Learning Model Optimization

2.3 Yolo8 Model Architecture

Ultralytics YOLOv8 model follows the classic Backbone -> Neck -> Head layout with some improvements from other YOLO architectures.

The key role of the Backbone is feature extraction. It converts the image into hierarchical feature maps (from low to high level).

Its key components are:

- Conv + BN + SiLU
- C2f blocks (main innovation)
- SPPF

The core idea of the C2f (Cross-Stage Partial with full feature fusion)

A bottleneck is a small stack of layers that:

1. Compresses channels(make tensor narrower)
2. Does some work (spatial mixing)

3. Expands channels back

It helps to drop redundant features, while preserving important.

SPPF = Spatial Pyramid Pooling – Fast

It's a context aggregation block used in YOLO (v5 $\rightarrow$ v8)

2.4 Performance Challenges in Machine Learning Deployment

Metrics:

- **Latency:** The time needed to process one inference request (etc. picture, text query);
- **Throughput:** The number of inference requests processed per unit of time (e.g., inferences per second).
- **Power Consumption:** The energy required to perform inference
- **Memory Requirements:** The amount of RAM and storage required for the model weights and intermediate activations

With the growing interest in on-device deployment, a clear gap has emerged between the training and inference stages of machine learning models. While large models are typically trained in data centers using powerful GPUs, inference is performed across a wide variety of deployment environments. Depending on the application, inference may be subject to strict constraints on memory, latency, and power consumption, and must run on heterogeneous hardware platforms such as CPUs, GPUs, TPUs, NPUs, or FPGAs. Each of these hardware architectures differs in structure, instruction sets, memory hierarchies, and supported data types. As a result, model performance depends heavily on how well it is adapted to the target hardware.

Machine learning models are usually developed in high-level Python frameworks such as TensorFlow or PyTorch. These frameworks often use dynamic execution (for example, PyTorch's eager mode), where:

- Operations are executed immediately, line by line
- Control flow (loops, conditionals) is resolved at runtime

- Tensors, layers, and operations are created and destroyed on the fly

That same flexibility comes at a cost:

- Python interpreter overhead Every operation goes through the Python interpreter, which is relatively slow.
- Interpreter locks (e.g., the GIL) Python’s Global Interpreter Lock can limit parallel execution.
- Object management Tensors and operators are Python objects that need to be created, tracked, and destroyed.
- Framework dispatch logic Each operation requires the framework to decide which kernel to run, on which device, and how to manage memory.

Individually, these costs are small — but in deep neural networks with thousands of operations, they add up.

In deployment, models are usually not run inside Python at all.

Instead, they are executed by:

- C++-based runtimes
- Hardware-specific inference engines
- Precompiled binaries on accelerators (GPU, NPU, TPU, FPGA)

These runtimes:

- Execute pre-compiled computation graphs
- Call highly optimized kernels directly
- Avoid Python entirely during inference

As a result, they remove most of the overhead associated with Python and dynamic execution.

Simply moving a model out of Python does not automatically guarantee speedups.

To fully benefit, the model must be:

- Compiled into a static or semi-static graph
- Optimized (operator fusion, memory planning, quantization, layout changes)
- Adapted to the target hardware’s instruction set and memory hierarchy

Without this careful compilation step, the deployed runtime may still execute inefficient kernels or suboptimal data flows.

Libraries like cuDNN and oneDNN are heavily optimized for throughput-oriented workloads, which historically means large batch sizes. Large batches increase data reuse (e.g., weights reused across many samples), which improves cache efficiency and amortizes memory access costs. With many parallel threads and large workloads, GPUs/CPU’s can overlap memory accesses with computation, effectively hiding memory latency. Many optimized kernels (e.g., GEMM-based convolutions) assume enough work to justify complex tiling, blocking, and scheduling strategies. These libraries were primarily developed to support large-scale training in data centers, where batch sizes of hundreds or thousands are common. So the default optimization targets of these libraries align better with large batches than with batch-1 inference, which is preferable for Real-time setups

Frameworks like PyTorch (in eager mode) execute operation by operation, where each op is dispatched independently; intermediate tensors are materialized in memory; the framework has limited visibility beyond the current operator. Compilers, in contrast, operate on a whole computation graph. All operators and data dependencies are visible; control flow and tensor lifetimes can be analyzed globally. Compilers merge multiple operations into a single kernel to reduce memory traffic and overhead.

With a full graph, compilers can:

- Remove redundant operations
- Fold constants
- Reorder associative/commutative ops
- Eliminate dead branches or unused tensors

Furthermore, different hardware prefers different layouts, GPUs often prefer NCHW, Some accelerators prefer NHWC or blocked layouts, so compilers can choose a single optimal layout, insert minimal layout conversions, avoid unnecessary transposes.

2.5 DNNs Compiler Workflow

An ML model compiler takes a *high-level model* (PyTorch / TensorFlow / ONNX) and turns it into efficient executable code for a *specific target* (CPU, GPU, NPU, FPGA, ASIC like Hailo-8).

Stages:

- Stage 1: Frontend (Model ingestion)
 - Input: PyTorch, TensorFlow, JAX
 - Exported formats: ONNX, TorchScript, StableHLO
 - Goal:
 - * Convert framework-specific graphs into a compiler IR
 - * Normalize control flow, ops, shapes
 - Key tasks:
 - * Canonicalize ops (e.g. many conv variants $\rightarrow$ one canonical conv)
- Stage 2: High-level graph optimizations (framework-agnostic)
- Stage 3: Lowering to tensor programs (IR $\rightarrow$ loops)
- Stage 4: Hardware-specific optimization & scheduling
- Stage 5: Code generation (backend)
- Stage 6: Runtime & execution

A typical ML execution flow involves translating a model defined in a high-level framework (like TensorFlow, PyTorch, or JAX) into a format suitable for high-performance execution on target hardware. This translation process is orchestrated by the ML compiler and runtime stack (see Fig. ??)

The Compiler’s primary responsibility is to convert high-level representation of a DNN model to an optimized lower-level executable by the target hardware.

Main stages:

- Conversion to an intermediate representation (IR);

- Graph level optimization: operation fusion and simplification, memory planning;
- Tensor/Loop-Level IR: tiling, vectorization, memory optimization;
- Code generation: instruction selection, register allocation, target-specific code emission;

ML runtime manages the dynamic aspects of model execution such as execution orchestration (loading compiled model artifact and managing its execution), memory management (allocation of memory for intermediate tensors), interfacing with hardware devices, task dispatching, and library integration.

Popular compilers and runtime stacks:

- TensorFlow/XLA (Accelerated Linear Algebra): compiles Tensorflow graphs via HLO IR;
- PyTorch 2.x (TorchDynamo, AOTAutograd, Inductor): captures Python bytecode into FX graph representation;
- Apache TVM: stack for optimizing models for CPU, GPU, microcontrollers, FPGAs and ASICs.
- MLIR-based Stacks (e.g., IREE, TensorFlow MLIR)
- ONNX Runtime

2.6 Performance Bottlenecks in ML Inference

A workload is compute-bound if increasing the raw processing power (e.g., higher clock speed, more compute units) directly leads to a proportional decrease in execution time, assuming memory and latency are not limiting factors.

Inference for complex neural networks requires millions of multiply-accumulate (MAC) operations. If hardware is not fully utilized and code not map vectors/tensors effectively, the processing bottlenecks are created.

Many ML operations are memory-bound. This means the execution time is dominated by the time spent transferring data between memory levels (e.g., DRAM to caches, cache levels, host-to-device memory) rather than the computation itself.

Latency bounded operations: single-batch inference, overhead for dispatching operations, pipeline stalls.

These strategies reduce dispatch overhead for frequently executed execution paths:

- **Ahead-of-Time (AOT):** Compile the entire model before execution, as in traditional C/C++ workflows and many edge deployment scenarios.
- **Just-In-Time (JIT):** Compile code or computation graphs at runtime, typically upon first execution, leveraging runtime information such as tensor shapes, data types, and target hardware.

2.7 Hailo Compiler

2.8 Methods for Pruning Deep Neural Network

In a survey [3] authors categorized over 150 studies of pruning neural networks. They chose 8 categories:

- **Sensitivity analysis methods (loss / gradient-aware pruning).** These methods are valuable as they prune what matters the most for the task and normally provide the highest accuracy-compression trade-off (50-90% accuracy with 1-2% accuracy drop).

Important works in this category:

- Taylor Pruning [4];
- SNIP (Single-shot Network Pruning) [5];
- GraSP (Gradient Signal Preservation) [6];
- Movement Pruning [7]

Although sensitivity-based pruning methods often achieve high unstructured sparsity (50–90%), such sparsity is difficult to exploit on general-purpose hardware due to irregular memory access and limited compiler support, resulting in limited real-world speedups. Consequently, structured pruning methods remain more practical for deployment across diverse hardware platforms.

- **Knowledge distillation methods (teacher-student compression).** Knowledge Distillation (KD) is a model compression paradigm in which a large, high-capacity model (Teacher) transfers knowledge to a smaller, more efficient model (Student). Instead of training the student solely on hard labels, KD trains it to match the teacher’s output behavior, typically using soft targets (probability distributions or logits). KD differs fundamentally from standard understanding of pruning: pruning removes parameters or structures, but KD transfers behavior. By doing this KD is normally does both: improves the accuracy by few percent

compared to the baseline and reduces the number of parameters up to 120 times. Important works:

- Knowledge Distillation [8];
- DCP [9];
- AutoCompress [10];

Knowledge distillation is a powerful model compression paradigm that transfers predictive behavior from a large teacher model to a compact student model using soft supervision. While classical distillation operates at the output level, recent works extend this idea to intermediate representations and pruning-aware settings. Methods such as discrimination-aware channel pruning integrate auxiliary losses to preserve discriminative power during structured pruning, achieving strong accuracy retention. However, distillation-based approaches are inherently task-dependent, require access to labeled data and a strong teacher, and introduce additional training complexity, limiting their applicability in representation-centric or hardware-agnostic compression scenarios.

- **Similarity and clustering methods** Similarity and clustering-based pruning methods aim to remove redundant filters or channels rather than filters that are individually “unimportant”. The core assumption is that modern CNNs are over-parameterized and contain filters whose functionality can be approximated or replaced by others with minimal impact on accuracy. Instead of ranking filters solely by magnitude or sensitivity, these methods explicitly model correlation, similarity, or representational overlap among filters. Representative methods include:
 - FilterSketch by Lin et al. [11],
 - Geometric Median Pruning by He et al. [12],
 - ThiNet by Lu et al. [13]
- **Low rank methods** are mostly theoretically grounded methods, effective in convolution-heavy models; Best methods: Show moderate speedups and faster accuracy degradation at high compression
- **Magnitude based pruning methods** are considered simple and cheap to implement models, however, they have limited compression and higher accuracy drops, compared to more complex structures; Best examples are L1 filter pruning (Li et al., 2017) [14], APoZ (Activation sparsity) [15].
- **Architectural design methods (NAS, RL search)**. examples: AMC, AutoSlim, Once-for-All (OFA)

- **Hybrid methods**
- **Quantization methods:** are not exactly pruning methods. They are usually combined with pruning. Mostly based on a transformation of weights from floating point to integer. Important works: Quantization-Aware Training (QAT) [16], Binary / Ternary Networks (XNOR, TWN) [17]

2.8.1 Magnitude based pruning

Magnitude based Pruning Methods are methods based on removing weights, reducing amount of inbound and outbound filters and nodes based on a measure of magnitude of the effect these filters have on the network. Pruning iteration is repeated until the accuracy is starting to drop. First works in this field were adopting the idea of removing the weight lower than a particular threshold.

Han et al. [18] carried out several experiments on LeNet-300-100 and Lenet-5, both trained on MNIST dataset, and showed that weight could be reduced by this method by a factor of 12 without significant drops in accuracy. Experiments based on AlexNet and VGG16 trained on ImageNet show the factor of 9 and 12 respectively. Han et al. note that L_1 regularization, a technique used in machine learning to prevent overfitting by adding a penalty term to the cost function, is better immediately after pruning, but that L_2 norm is better, when the weights of the pruned model are fine-tuned. Experiments also suggest that earlier layers are the most sensitive to pruning and that iterative pruning is better than one-shot pruning (cutting the proportion of weights in one cycle). Another observation from their work is that re-initialization of the weights is not enough to construct an accurate model. It is always better to fine-tune the weights of the pruned model.

Later work by Guo et al. [19] notes that magnitude pruning can lead to premature removal of weight that can become important given removal of other weights. They propose a method called Dynamic Network Surgery to keep track of pruned weights and restore them if they turn to be important. For AlexNet on ImageNet they reach a factor of 17 without compromising accuracy.

The other method proposed by the Frankle and Carbin (2018) is the lottery ticket hypothesis proposed by [20] and further continued by Frankle et al. (2019) in [21]. It formulates a statement, that a trained network contains a subnetwork, which can be trained to be at least as accurate as the original network using no more than the number of epochs used for training the original network. This subnetwork is a so-called winning

lottery ticket. Authors test two pruning methods. The first is a magnitude pruning ($p\%$ of the weights), after which the remaining weights are reset to their initial values and are retrained. The second method is using an alternative pruning to iteratively prune $p^{\frac{1}{n}}$ of the weights in n cycles. They experimented on VGG and ResNet and on some bigger models suggested that setting of the weights to those obtained from later iteration of training can be beneficial when high levels of pruning are used (more than 30%).

Another question arises is if these subnetworks are transferable to other tasks. Hubens et al. (2020) [22] show that when a network is trained on a larger data set, such as ImageNet, and transferred and fine-tuned for a different task, pruning can result in a smaller network than if it was trained from scratch on the new task.

Despite the experiments proving that some unstructured pruning methods allow a significant drop of weights, not all hardware can process sparse weight matrices and support speed up inference. Pruning of higher granularity structures (filters and channels) or so-called structured pruning helps to avoid this problem, as many existing toolkits support it. Survey [3] suggests three main directions of structured pruning research:

- **Data dependent channel pruning methods.** A useful channel should respond differently to different inputs. A channel that barely changes for any input is potentially useless.

Polyak & Wolf (2015) [23] propose two methods - inbound pruning (to reduce the amount of input channels) and “Reduce and Reuse” pruning (to cut output channels).

The assessment of input channels is performed by applying the network to a sample of images and measuring the variance in a feature map. These measures are ranked and those below a threshold are pruned.

The “Reduce and Reuse” method targets a reduction in the number of output channels produced by a convolutional layer. The underlying assumption is that output channels exhibiting low variation are less informative and can therefore be removed with minimal impact on performance. A key challenge of pruning output channels is that each removed channel is expected as an input by subsequent layers. To address this issue, the Reduce and Reuse method approximates the removed channels as linear combinations of the remaining ones. This approximation is obtained by solving a least-squares optimization problem that minimizes the reconstruction error between the original and approximated outputs. The resulting transformation is implemented as an additional 1×1 convolutional layer, allowing the network to preserve representational capacity while reducing the number of channels.

The results from their experiments show that the variance based method is more

effective than use of random pruning.

- **Data independent pruning methods.** Relay on the assumption that properties of the filters and output channels, as the proportion of zeros, influence the structure’s importance.

Hu et al. (2016) [15] propose APoZ-based network trimming, a pruning strategy that does not rely on data-driven measures such as feature map variance or entropy. Instead, their method is based on the observation that a large number of zero activations in an output feature map indicates redundancy. Specifically, they introduce the Average Percentage of Zeros (APoZ) as a metric to quantify the weakness of a filter, with higher APoZ values suggesting lower importance. After each pruning step, the network is fine-tuned, and the authors report that fine-tuning yields better results than retraining the network from scratch. When evaluating the proposed method on VGG-16, the authors achieved a compression rate of 2.59 after six pruning iterations, while 2-3% higher Top-1/Top-5 accuracy than the original VGG-16 model.

- **Optimization based channel approximation pruning methods** propose to remove some channels and then re-optimize the remaining ones instead of deciding which channels are important, so that the layer behaves the same.

One representative approach is FilterSketch, proposed by Lin et al. (2021) in [11]. Experimental results demonstrate that FilterSketch consistently outperforms first-order magnitude-based pruning methods, FilterSketch removes around 41.5% of the FLOPs and parameters while keeping the top-1 accuracy at 93.19%. Compared to 93.26% by the pre-trained model, the accuracy drop is negligible on CIFAR-10.

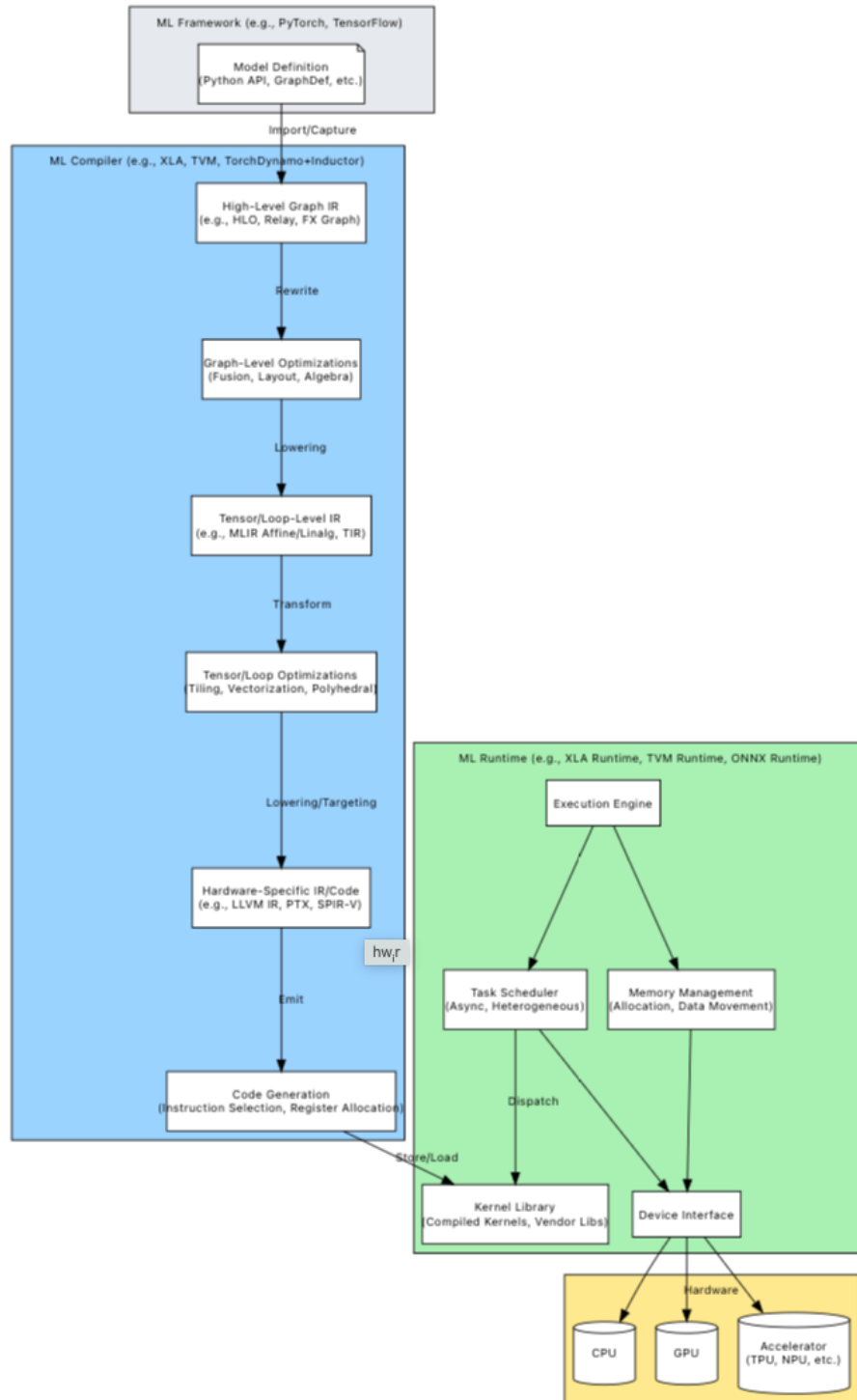


Figure 2.1: Flow of an ML model from definition through compilation and runtime execution on hardware.

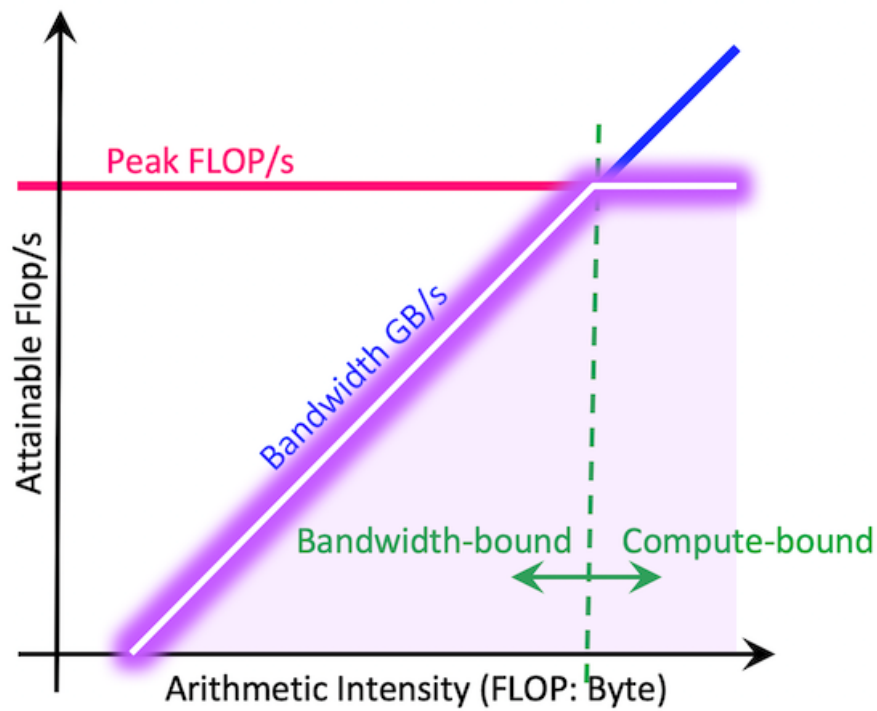


Figure 2.2: A Roofline model illustrating performance limits based on compute capability (horizontal line) and memory bandwidth (sloped line) versus the arithmetic intensity of operations (dots). Operations below the "roof" are limited by either memory bandwidth or compute peak.

3 Theory

4 Methodology

5 Proposed Solution / Design

5.1 Roofline model

The Roofline model is a visual performance model that relates a program's achievable performance to the hardware's peak compute capability and memory bandwidth. It plots performance (e.g., GFLOP/s) versus arithmetic intensity (FLOPs per byte of memory traffic). The model defines two limits: a sloped memory roof determined by peak memory bandwidth, and a flat compute roof determined by peak floating-point throughput. Any kernel's performance is bounded by the minimum of these two ceilings. If a point lies on the sloped region, it is memory-bound; if it lies on the flat region, it is compute-bound. The Roofline model is widely used to analyze whether a workload is limited by memory movement or computation and to guide optimization efforts.

Optimization strategy of a neural network model is dependent on whether our device limits us in the speed of reading/writing to/from memory or the computational speed overall.

- **If the workload is Memory-Bound:**

This means performance is limited by memory bandwidth, not by compute units. The arithmetic intensity (FLOPs/byte) is low, so the kernel spends most of its time moving data. The goal is then to either increase arithmetic intensity, or reduce memory traffic.

Optimization strategies:

- Improve data locality: reuse data in caches (tiling/blocking); Keep tensors in shared memory (GPU);
- Fuse operations to avoid writing intermediate tensors to memory (Conv → BatchNorm → Silu) → (Fused Conv-BN-Activation)
- Reduce Precision (FP32 → FP16 → INT8 → INT4)
- Reduce tensor size: pruning, structured sparsity, channel reduction, smaller input resolution
- Improve memory layout: avoid unnecessary copies, channel-last format (NHWC)

- **If the workload is Compute-Bound:**

This means arithmetic units are saturated. Arithmetic intensity is high enough that

bandwidth is no longer limiting. The goal is then to increase compute efficiency. This case is harder, as it requires the work with scheduling, parallelization, kernel optimization and algorithmic improvements.

Official Yolov8 parameters are:

Table 5.1: Comparison of YOLO models (trained on COCO) at 640×640 resolution. [24]

Model	CPU ONNX (ms)	A100 TensorRT (ms)	Params (M)	FLOPs (B)
v8n	80.4	0.99	3.2	8.7
v8s	128.4	1.20	11.2	28.6
v8m	234.7	1.83	25.9	78.9
v8l	375.2	2.39	43.7	165.2
v8x	479.1	3.53	68.2	257.8

Macbook Pro M3 has declared 150GB/s memory bandwidth [25]. Raspberry Pi 5 has up to 17 GB/s of memory bandwidth [26], NVIDIA’s latest software update enhances Jetson Orin Nano’s AI performance from 40 to 67 AI TOPs and *memory bandwidth from 68 to 102 GB/s* [27].

Raspberry Pi 5 processor [26] is a broadcom BCM2712 2.4GHz quad-core 64-bit Arm Cortex-A76 CPU, with Cryptographic Extension, 512KB per-core L2 caches, and a 2MB shared L3 cache

Calculating the statistics of roofline model can be tricky. First, there is a question, how different are the documented by manufacturer statistics from the observed ones. Second, how a particular model would be run on this hardware, how effective is the use of cache already, would all intermediate vectors be stored in the memory or there are some hidden optimizations (fused layers).

- **Peak computational performance (in GFLOP/s)** is obtained by benchmarking a compute-bound kernel that saturates the floating-point execution units. In practice, this can be achieved using large dense matrix multiplications of size $m \times k$ and $k \times n$, for which the floating-point operation count is given by:

$$\text{FLOPs} = 2 * m * k * n. \quad (5.1)$$

The sustained peak performance is then computed as

$$P_{\text{peak}} = \frac{\text{FLOPs}}{t_{\text{compute}}}, \quad (5.2)$$

where t_{compute} denotes the measured execution time.

- **Peak memory bandwidth (in GB/s)** is determined using a streaming memory benchmark. A large device-to-device copy or STREAM-like triad kernel is executed over a sufficiently large buffer to avoid cache effects. The sustained memory bandwidth is computed as

$$B_{\text{peak}} = \frac{\text{Bytes transferred}}{t_{\text{memory}}}. \quad (5.3)$$

- **Arithmetic intensity (AI)** is defined as the ratio between the number of floating-point operations and the volume of data transferred to and from main memory:

$$\text{AI} = \frac{\text{FLOPs}}{\text{Bytes}}. \quad (5.4)$$

Since exact DRAM traffic is generally unavailable without hardware performance counters, a lower-bound estimate of memory traffic is commonly used. For convolutional or linear layers, this lower bound consists of the input tensor, weight parameters, and output tensor:

$$\text{Bytes}_{\text{LB}} = \text{Bytes}_{\text{input}} + \text{Bytes}_{\text{weights}} + \text{Bytes}_{\text{output}}. \quad (5.5)$$

This estimate ignores cache reuse and intermediate buffering, thereby providing an upper bound on arithmetic intensity.

- **The achieved performance of a workload** is computed as

$$P_{\text{achieved}} = \frac{\text{FLOPs}}{t_{\text{execution}}}. \quad (5.6)$$

The Roofline bound is then defined as

$$P_{\text{roof}}(\text{AI}) = \min(P_{\text{peak}}, B_{\text{peak}} \times \text{AI}). \quad (5.7)$$

If P_{achieved} approaches $B_{\text{peak}} \times \text{AI}$, the workload is memory-bound. Conversely, if it approaches P_{peak} , the workload is compute-bound. This methodology enables systematic classification of performance bottlenecks and guides optimization strategies such as improving data locality, increasing arithmetic intensity, or enhancing computational parallelism.

Table 5.2 summarizes representative per-layer performance characteristics of YOLOv8n executed on the Raspberry Pi 5 CPU. The measured sustained memory bandwidth was 9.14 GB/s and peak compute throughput was 69.86 GFLOP/s. The roofline knee occurs at approximately 8.1 FLOPs/byte, separating memory-bound from compute-bound regimes.

Layers with low arithmetic intensity ($AI \approx 7-8$) achieve limited performance (3–6 GFLOP/s), indicating memory-bound behavior. In contrast, deeper convolutional layers with high arithmetic intensity ($AI > 70$ FLOPs/byte) achieve 40–60 GFLOP/s, approaching 70–85% of measured peak compute throughput, demonstrating compute-bound execution.

The detection head’s distribution focal loss (DFL) convolution exhibits extremely low arithmetic intensity (0.47 FLOPs/byte) and achieves only 0.4 GFLOP/s, confirming strong memory limitation.

Table 5.2 shows per-layer and overall performance statistics of YOLOv8n on raspberry Pi 5 (CPU, FP32). Although several convolutional layers individually approach peak compute throughput, the full model achieves only 12.36 GFLOP/s due to additional non-convolutional operations, synchronization overhead, framework dispatch costs, and inter-layer memory traffic. Overall, YOLOv8n inference on Raspberry Pi 5 is predominantly compute-bound for high-intensity convolutional blocks, while early feature extraction and lightweight layers remain memory-bound.

Table 5.2: Per-layer performance statistics of YOLOv8n on Raspberry Pi 5 (CPU, FP32).

Layer	Time (ms)	AI (FLOPs/byte)	Perf (GFLOP/s)	Regime
0.conv	23.79	7.71	3.7	Memory-bound
2.cv1.conv	8.88	7.99	5.9	Memory-bound
4.m.0.cv2.conv	2.82	70.42	41.8	Compute-bound
6.m.1.cv2.conv	2.06	122.03	57.3	Compute-bound
18.m.0.cv2.conv	1.95	122.03	60.5	Compute-bound
22.cv3.0.1.conv	13.86	170.41	53.2	Compute-bound
22.dfl.conv	2.42	0.47	0.4	Strongly memory-bound
Full Model	358.35	157.76	12.36	Mixed / Overhead-dominated

The roofline analysis shows that YOLOv8n on the Raspberry Pi 5 operates predominantly in the compute-bound regime for high-arithmetic-intensity convolutional layers, while a few early and lightweight layers remain memory-bound. Although several individual convolutional blocks approach the measured peak compute throughput (≈ 70 GFLOP/s), the full model achieves only 12.36 GFLOP/s, corresponding to roughly 17% of peak performance.

This gap indicates that overall inference performance is not limited by raw memory

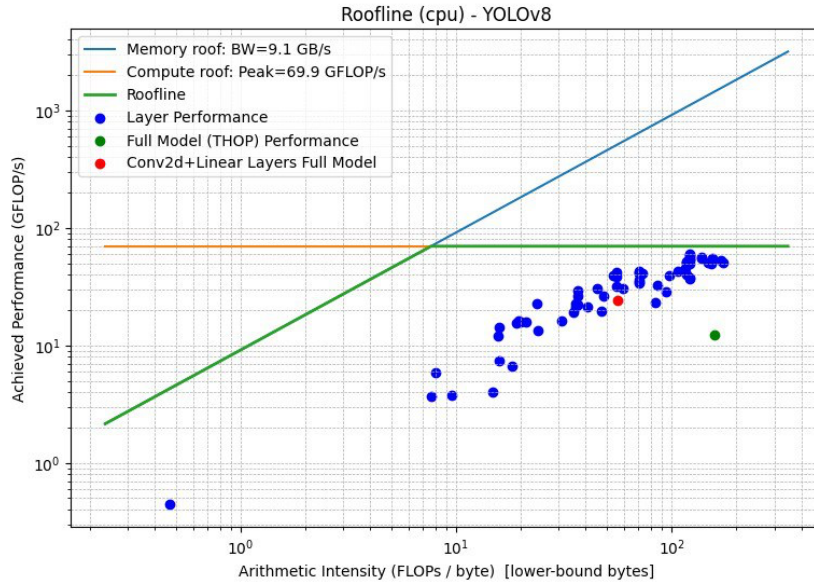


Figure 5.1: Roofline model of YOLOv8n on Raspberry Pi 5 (CPU, FP32).

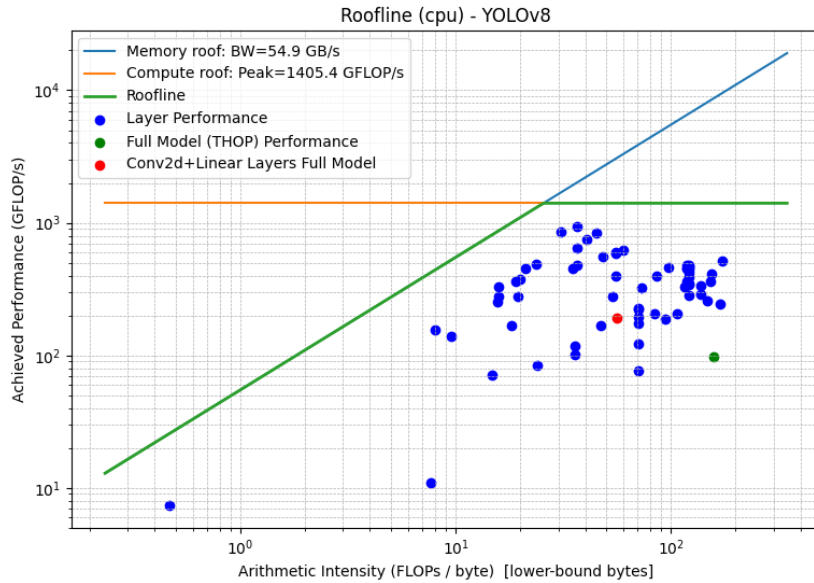


Figure 5.2: Roofline model of YOLOv8n on Macbook Pro M3 (CPU, FP32).

bandwidth but rather by computational inefficiencies and system-level overhead, including kernel launch costs, framework dispatch, synchronization, non-convolutional operations, and inter-layer memory traffic. The roofline therefore does not represent an achievable target, but an upper theoretical bound.

From a research perspective, the results suggest that further optimization should focus on improving computational efficiency through kernel fusion, graph compilation, operator reduction, and quantization, rather than purely reducing memory traffic.

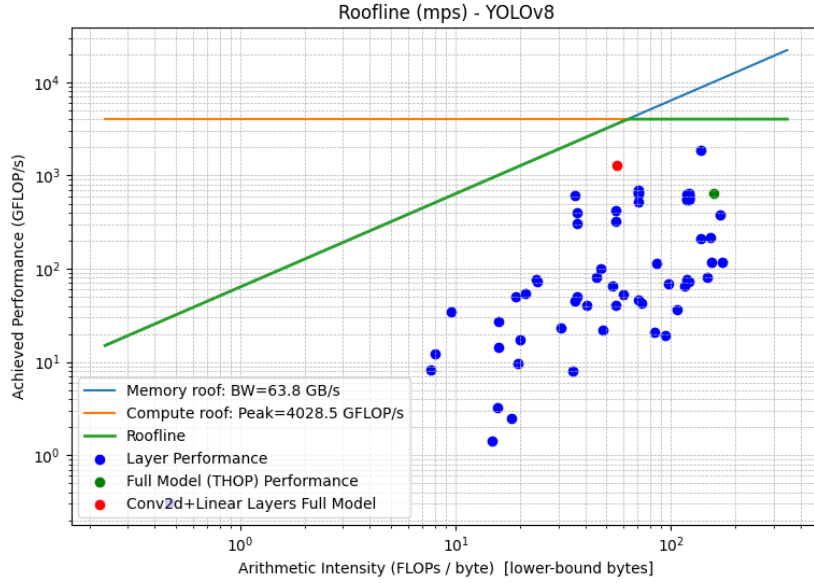


Figure 5.3: Roofline model of YOLOv8n on Macbook Pro M3 (MPS, FP32).

Raspberry Pi 5 cache hierarchy is:

- L1d: 256 KiB (4 instances) = 64 KB per core
- L1i: 256 KiB (4 instances) = 64 KB per core
- L2: 2 MiB (4 instances) = 512 KB L2 per core → small
- L3: 2 MiB (1 instance) → shared last-level cache
- DRAM bandwidth ≈ 9 GB/s (measured)

So layers larger than 2 MB activation footprint will spill to DRAM.

So, early YOLO layers (e.g., 320×320 feature maps) cannot fit in cache. Later layers (40×40 , 20×20) may fit in L3 → more compute-bound behavior.

5.2 GPU structure

In this work we use NVIDIA Orin Nano Platform. It is an embedded AI computing module designed for edge inference and robotics applications. Its SoC combines: ARM-based CPU with 6 Cores, NVIDIA Ampere-based GPU and various I/O. The shared by CPU and GPU LPDDR memory reduces latency and speeds-up edge-workloads. The GPU in Orin Nano supports CUDA, Tensor Cores, Mixed precision (FP32, FP16, INT8).

5.3 Apache TVM

Apache TVM is a modular ML compiler framework that transforms high-level tensor programs into optimized low-level code and runtime artifacts. It is an Open Source Project, mainly developed by a Tianqi Chen’s team at Carnegie Mellon University.

Deep learning frameworks make models easy to define, train, and export. However, they do not always produce hardware-efficient binaries for every deployment target. TVM exists to close that gap: it turns model programs into optimized executables across heterogeneous hardware. It also provides deep optimization control over the model.

TVM addresses this with a compiler-first approach: represents computation in intermediate forms, applies transformations, searches optimization spaces, and generates code for many targets.

Core architecture

TVM can be understood as a pipeline:

1. Frontend import: ONNX, PyTorch-exported graphs, TensorFlow, etc.
2. Graph-level IR transformations: operator fusion, layout rewrite, constant folding.
3. Tensor-level lowering and scheduling: loop tiling, vectorization, unrolling, memory placement.
4. Target code generation: LLVM for CPU, CUDA for NVIDIA GPU, other backends.
5. Runtime packaging: graph executor, VM, or AOT runtime.

```
1 def add(a, b):  
2     return a + b
```

TVM performs hardware-aware graph partitioning.

5.4 Dataset and Annotation

6 Testing and Results

6.1 Experimental Setup

6.1.1 Performance overview

6.1.2 Prediction Examples

6.1.3 Hailo impact

6.2 FP32 Runtime Comparison on Jetson Orin

To establish a consistent FP32 baseline across deployment runtimes, a unified benchmark was executed on Jetson Orin on March 13, 2026 (2026-03-13 15:27:50). The run was performed with the `bench` command of the shared evaluation pipeline, which measures four inference backends under the same input tensor shape and benchmarking procedure:

- TVM (Relax VM, loaded from a precompiled `.so`),
- ONNX Runtime CUDA Execution Provider,
- ONNX Runtime TensorRT Execution Provider,
- PyTorch (Ultralytics model execution).

The configuration (`gpu_fp32.yaml`) used:

- input model: `yolov8n_opset17_fp32.onnx`,
- PyTorch checkpoint: `yolov8n.pt`,
- device: CUDA (`cuda:0`),
- precision: FP32,
- input size: 640×640 , batch size = 1,
- benchmark protocol: 200 measured runs with 20 warmup iterations.

For TVM, the evaluated module was compiled using an existing MetaSchedule database generated in an earlier tuning stage. The tuning logs indicate a substantial search effort (4282 total trials), suggesting that the measured TVM result reflects a tuned, deployment-oriented build rather than an untuned baseline.

Runtime	Latency [ms]	FPS
ONNX Runtime TensorRT EP	13.5292	73.9142
TVM (Relax VM)	23.5091	42.5368
ONNX Runtime CUDA EP	23.8097	41.9998
PyTorch (Ultralytics)	24.7877	40.3426

Table 6.1: FP32 runtime benchmark results on Jetson Orin (640×640 , batch = 1).

The results show a clear separation between TensorRT-backed and non-TensorRT execution in this setup. ONNX Runtime with TensorRT EP achieved the lowest latency (13.53 ms), corresponding to 73.91 FPS, while TVM, ONNX Runtime CUDA EP, and PyTorch formed a second cluster around 23.5–24.8 ms (~ 40 –43 FPS). Relative to ONNX Runtime CUDA EP, TensorRT EP reduced latency by approximately 43.2% (about $1.76\times$ speedup). TVM and ONNX Runtime CUDA EP were close in performance, with TVM slightly faster in this run.

Overall, this experiment provides a practical FP32 deployment baseline for subsequent quantization analysis. It also indicates that, on Jetson Orin, backend selection can have an impact comparable to model-level optimization: the same FP32 network exhibits markedly different inference speed depending on the runtime execution provider.

6.3 Quantization Influence on YOLOv8n

Statistical significance was evaluated with a paired bootstrap test on validation images (1000 resamples), and confidence intervals were computed using the percentile method. The results indicate that INT8 quantization causes a statistically significant decrease in detection accuracy.

6.3.1 TensorRT FP32 vs INT8 Benchmarking on Jetson Orin

To compare TensorRT performance for FP32 and INT8 inference on Jetson Orin, two engines were exported from the same YOLOv8n model (FP32 and INT8) using Ultralytics export. Benchmarks used batch size 1 and input size 640×640 . Reported latency is end-to-end per-batch latency, including host invocation, device execution, synchronization, and output transfer.

Export parameters.

- Model: `yolov8n.pt`
- Input: 640×640 , batch size = 1
- Device: GPU (`device=0`)
- TensorRT workspace: 4 GB
- FP32 engine: `int8=False, half=False`
- INT8 engine: `int8=True, half=False, data=DATA_YAML, fraction=1.0, split="train"`

Early measurements showed strong sensitivity to benchmarking procedure. In particular, when both engines were loaded and benchmarked sequentially in one process, the second run often benefited from a warmer runtime state. To control this effect, additional experiments were performed with:

- reversed engine order,
- longer warmup (500 iterations),
- fixed Jetson performance settings (`sudo nvpmodel -m 0, sudo jetson_clocks`).

Phase	Run mode	FP32 ms	FP32 FPS	INT8 ms	INT8 FPS	INT8 gain
No clocks	dual (int8→fp32)	21.274	47.006	20.473	48.844	3.8%
No clocks	dual (fp32→int8)	21.067	47.469	15.545	64.330	26.2%
No clocks	separate runs	23.122	43.248	20.249	49.385	12.4%
With clocks	dual (int8→fp32)	16.459	60.756	9.876	101.252	40.0%
With clocks	dual (fp32→int8)	16.768	59.639	9.655	103.569	42.4%
With clocks	separate runs	16.588	60.284	10.068	99.328	39.3%

Table 6.2: TensorRT YOLOv8n benchmark summary (batch=1, runs=1000, warmup=500).

Under locked-performance conditions, the INT8 engine achieved approximately $1.65\times$ higher throughput than FP32. Before locking power and clocks, the measured FP32–INT8 gap varied substantially with run order and process isolation. After enabling maximum performance mode and fixed clocks, measurements became more stable and the INT8 advantage increased. The resulting measurements are summarized in Table 6.2.

This pattern suggests that earlier variability was dominated by DVFS and runtime warmup effects. In sequential same-process benchmarking, the later run can benefit from an already initialized CUDA runtime, warmed TensorRT execution context, and elevated GPU/memory clocks. These effects are especially pronounced on Jetson platforms.

Therefore, Jetson benchmarking should report not only numerical performance but also measurement protocol, warmup configuration, and device power state.

`sudo nvpmodel -m 0` selects a high-performance power mode (power budget, online cores, and allowed CPU/GPU/EMC clock limits). `sudo jetson_clocks` then fixes clocks near their maximum allowed values.

Although these settings improve peak performance, they are not always appropriate for continuous use:

- higher power consumption,
- increased heat and possible thermal throttling in long runs,
- more fan noise,
- lower realism for power-managed deployment scenarios,
- reduced responsiveness for mixed workloads,
- greater long-term thermal stress.

A practical approach is to enable them for controlled benchmarking and disable them for routine development or power-constrained deployment.

6.4 YOLO pruning

The main caveat is that once YOLOv8n is converted into a TensorRT engine, TensorRT may fuse layers, insert reformat, and choose different kernels/tactics. So the “slow part” you see in TensorRT is not always a 1:1 match with the original YOLO PyTorch layers. NVIDIA explicitly notes that each TensorRT layer can launch multiple kernels, and extra reformat operations may appear as kernels or device-to-device copies.

7 Discussion

7.1 Discussion Points

7.1.1 Quantization and accelerator deployment

7.1.2 Unexplored hardware configurations and deployment possibilities

8 Conclusion

9 Future Work

10 Description of Use of Artificial Intelligence

In working on this specialization project, I have used artificial intelligence (AI) tools in the following limited ways. I used ChatGPT (OpenAI) to assist with idea development, clarifying concepts, and improving the structure and language of the text (for example, by suggesting alternative formulations, checking grammar, and helping to simplify explanations). AI was also used to get help with technical formatting issues in $\text{\LaTeX}$ and to debug small code examples by explaining error messages and proposing corrections.

All scientific content, including the problem formulation, choice of methods, implementation, experiments, analysis, and conclusions, has been developed, reviewed, and academically assessed by me. AI tools were not used to generate entire sections of the report, to perform literature searches independently, or to answer the assignment in its entirety. I have not uploaded sensitive or confidential information to the AI tools. I am solely responsible for the academic content of this assignment.

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A Demonstration (Appendix A)

Table A.1: Per-layer performance statistics of YOLOv8n on Raspberry Pi 5 (CPU, FP32). Regime is derived from the roofline knee $AI_{\text{knee}} \approx 7.64$ FLOPs/byte (using measured BW=9.14 GB/s and peak=69.86 GFLOP/s): Memory-bound if $AI < 6.88$, Knee/transition if $6.88 \leq AI \leq 8.40$, Compute-bound if $AI > 8.40$.

Layer	Time (ms)	AI (FLOPs/byte)	Perf (GFLOP/s)	Regime
model.0.conv	23.79	7.713	3.7	Knee / tion
model.1.conv	17.59	23.955	13.4	Comput bound
model.2.cv1.conv	8.88	7.995	5.9	Knee / tion
model.2.m.0.cv1.conv	5.17	35.899	22.8	Comput bound
model.2.m.0.cv2.conv	5.53	35.899	21.3	Comput bound
model.2.cv2.conv	21.02	9.593	3.7	Comput bound
model.3.conv	11.98	47.291	19.7	Comput bound
model.4.cv1.conv	7.08	15.920	7.4	Comput bound
model.4.m.0.cv1.conv	3.45	70.416	34.1	Comput bound
model.4.m.0.cv2.conv	2.82	70.416	41.8	Comput bound
model.4.m.1.cv1.conv	3.28	70.416	36.0	Comput bound
model.4.m.1.cv2.conv	3.43	70.416	34.4	Comput bound
model.4.cv2.conv	6.62	21.192	15.8	Comput bound
model.5.conv	7.30	85.714	32.3	Comput bound

Continued on n

Layer	Time (ms)	AI (FLOPs/byte)	Perf (GFLOP/s)	Regime
model.6.cv1.conv	3.24	30.769	16.2	Comput bound
model.6.m.0.cv1.conv	3.18	122.034	37.1	Comput bound
model.6.m.0.cv2.conv	3.13	122.034	37.7	Comput bound
model.6.m.1.cv1.conv	2.40	122.034	49.2	Comput bound
model.6.m.1.cv2.conv	2.06	122.034	57.3	Comput bound
model.6.cv2.conv	4.91	40.506	21.4	Comput bound
model.7.conv	5.96	97.959	39.6	Comput bound
model.8.cv1.conv	1.98	48.485	26.5	Comput bound
model.8.m.0.cv1.conv	2.92	118.033	40.4	Comput bound
model.8.m.0.cv2.conv	2.27	118.033	52.0	Comput bound
model.8.cv2.conv	2.46	55.491	31.9	Comput bound
model.9.cv1.conv	1.36	35.165	19.2	Comput bound
model.9.cv2.conv	3.42	59.813	30.7	Comput bound
model.12.cv1.conv	5.18	45.283	30.4	Comput bound
model.12.m.0.cv1.conv	2.22	122.034	53.1	Comput bound
model.12.m.0.cv2.conv	2.01	122.034	58.6	Comput bound
model.12.cv2.conv	3.56	36.641	22.1	Comput bound
model.15.cv1.conv	6.91	23.821	22.8	Comput bound

Continued on n

Layer	Time (ms)	AI (FLOPs/byte)	Perf (GFLOP/s)	Regime
model.15.m.0.cv1.conv	2.90	70.416	40.6	Comput bound
model.15.m.0.cv2.conv	2.76	70.416	42.8	Comput bound
model.15.cv2.conv	5.09	19.085	15.4	Comput bound
model.16.conv	3.02	53.731	39.1	Comput bound
model.18.cv1.conv	2.96	36.641	26.5	Comput bound
model.18.m.0.cv1.conv	2.19	122.034	53.9	Comput bound
model.18.m.0.cv2.conv	1.95	122.034	60.5	Comput bound
model.18.cv2.conv	2.70	36.641	29.1	Comput bound
model.19.conv	2.89	73.096	40.8	Comput bound
model.21.cv1.conv	2.03	55.491	38.8	Comput bound
model.21.m.0.cv1.conv	2.35	118.033	50.2	Comput bound
model.21.m.0.cv2.conv	2.32	118.033	50.7	Comput bound
model.21.cv2.conv	1.87	55.491	42.0	Comput bound
model.22.cv2.0.0.conv	8.48	137.799	55.7	Comput bound
model.22.cv2.0.1.conv	8.35	137.799	56.5	Comput bound
model.22.cv2.0.2	3.71	15.919	14.1	Comput bound
model.22.cv2.1.0.conv	4.30	154.839	54.9	Comput bound
model.22.cv2.1.1.conv	2.14	122.034	55.2	Comput bound

Continued on n

Layer	Time (ms)	AI (FLOPs/byte)	Perf (GFLOP/s)	Regime
model.22.cv2.1.2	1.08	15.681	12.1	Comput bound
model.22.cv2.2.0.conv	2.78	107.063	42.4	Comput bound
model.22.cv2.2.1.conv	1.28	83.721	23.1	Comput bound
model.22.cv2.2.2	0.83	14.798	4.0	Comput bound
model.22.cv3.0.0.conv	11.92	152.381	49.5	Comput bound
model.22.cv3.0.1.conv	13.86	170.414	53.2	Comput bound
model.22.cv3.0.2	5.15	19.874	15.9	Comput bound
model.22.cv3.1.0.conv	5.81	173.494	50.8	Comput bound
model.22.cv3.1.1.conv	3.65	146.939	50.4	Comput bound
model.22.cv3.1.2	1.27	19.506	16.2	Comput bound
model.22.cv3.2.0.conv	3.24	115.663	45.5	Comput bound
model.22.cv3.2.1.conv	1.60	94.737	28.7	Comput bound
model.22.cv3.2.2	0.76	18.161	6.7	Comput bound
model.22.dfl.conv	2.42	0.471	0.4	Strongly memory
Full Model (THOP)	358.35	157.765	12.36	Mixed Overhea dominat

B Demonstration (Appendix B)

YOLOv8n summary: 129 layers, 3,157,200 parameters, 0 gradients, 8.9 GFLOPs.

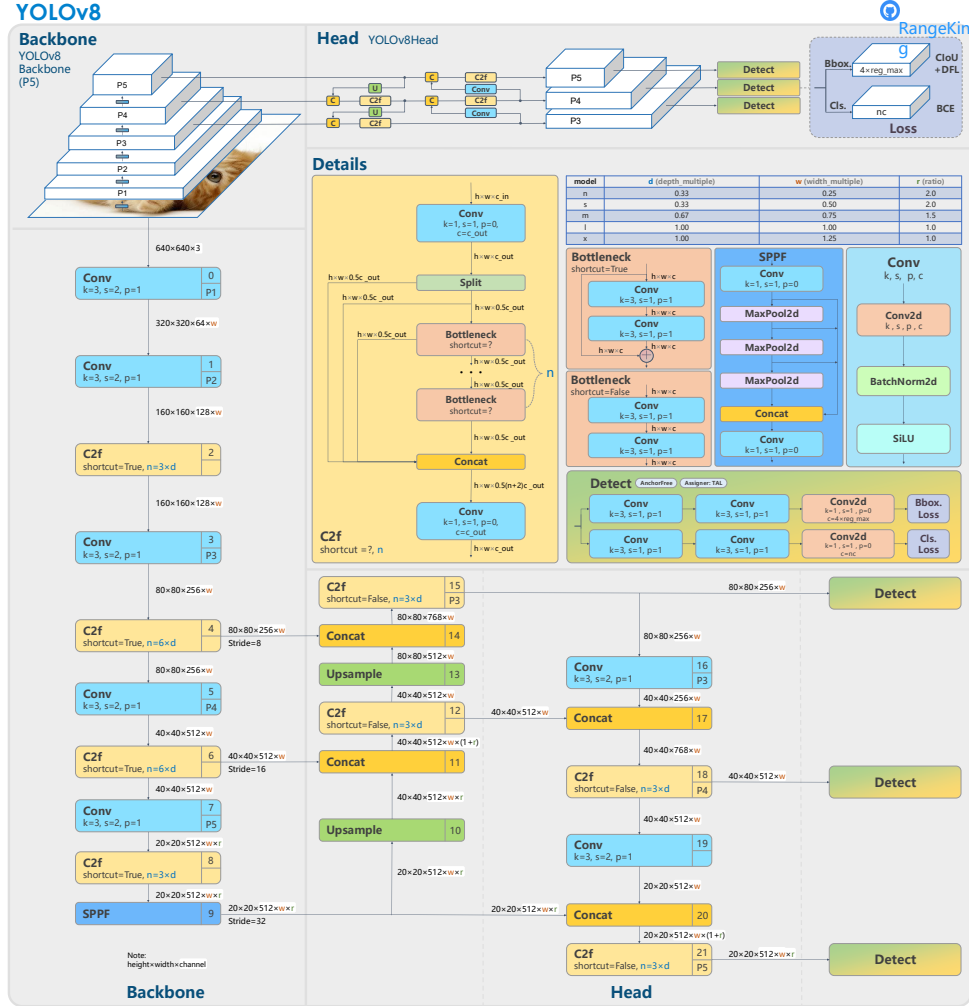


Figure B.1: YOLOv8 structure [28]